

CS 105: Department Introductory Course on Discrete Structures

Instructor : S. Akshay

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Lecture 06 – Basic structures: sets and functions

Recap of last 5 lectures

Week 01 and Week 02

- ▶ What are discrete structures?
- ▶ Chapter 1: Proofs and mathematical reasoning
 - ▶ Propositions: statements that can be assigned a truth value
 - ▶ Predicates and Quantifiers
 - ▶ Theorems and types of proofs: contradiction, contrapositive

Week 02 and Week 03

- ▶ Induction
- ▶ Well-ordering principle
- ▶ Strong Induction

Three problem-sheets released on Piazza

1. Questions on Basic proofs, reasoning
2. Questions on Induction, WOP, Strong Induction.

Logistics

Optional Problem Solving Sessions started...

- ▶ From now on at **3.30pm Wednesdays** (Same venue: CC101/103/105)
- ▶ Tomorrow, pop-quiz answersheets will be shown in **first 15min**.
- ▶ Check Piazza before coming!

Safe and Attendance

- ▶ Any remaining issues: Private message on Piazza to Instructors with proof (screenshots etc).
- ▶ Attendance $\neq$ Presence!

Pop Quiz Anytime!

Quiz 1: Aug 28th, 3.45pm NOTE THE CHANGE!

From proofs to structures

Next: Chapter 2: Basic Discrete Structures

- ▶ Finite and Infinite Sets,
- ▶ Functions
- ▶ Relations

Sets



§ 1
The Conception of Power or Cardinal Number
By an “aggregate” (*Menge*) we are to understand any collection into a whole (*Zusammenfassung zu einem Ganzen*) M of definite and separate objects m of our intuition or our thought. These objects are called the “elements” of M .

Figure: Georg Cantor (1845-1918); extract

What is a set?

- ▶ A **set** is an unordered collection of objects.
- ▶ The objects in a set are called its **elements**.

More formally,

Let P be a property. Any collection of objects that are defined by (or satisfy) P is a set, i.e., $S = \{x \mid P(x)\}$.

Some simple boring stuff about sets

Examples and properties

- ▶ We have already seen examples: $\mathbb{Z}, \mathbb{N}, \mathbb{R}$, set of all horses,...
- ▶ Let A, B be two sets. Recall the usual definitions:
 - ▶ Equality $A = B$, Subset $A \subseteq B$,
 - ▶ Cartesian product $A \times B = \{(a, b) \mid a \in A, b \in B\}$
 - ▶ Union $A \cup B = \{x \mid a \in A \text{ or } b \in B\}$
 - ▶ Intersection $A \cap B = \{x \mid a \in A \text{ and } b \in B\}$
 - ▶ Empty set ϕ ,
 - ▶ Power set of $A = \mathcal{P}(A)$ = set of all subsets of A .
 - ▶ If U is the universe, then the complement of A , $\bar{A} = A^c = \{x \in U \mid x \notin A\}$.

So, what is the difference between $\{\emptyset\}$ and $\emptyset$?

Not so simple...

A barber is a man in town who only shaves those who don't shave themselves.

Barber's paradox: Does the barber shave himself?

Russell's paradox

Let $S = \{X \mid X \notin X\}$

Then if $S \in S$, then $S \notin S$ and if $S \notin S$, then $S \in S$!

Axiomatic approach to set theory (ZFC!)

Start with a few objects **defined**. Then for a set A and a property P ,

$S = \{x \in A \mid P(x)\}$ is a set.

Let $P(x) = x \notin x$. let A be a set and $S = \{x \in A \mid x \notin x\}$.

- ▶ if $(S \in S)$: from the definition of S , $S \in A$ and $S \notin S$, which is a contradiction.
- ▶ if $(S \notin S)$: from the definition, either $S \notin A$ or $S \in S$. But we have assumed that $S \notin S$. Hence, $S \notin A$. No contradiction!

Is the fun done? How do we compare sets?

- ▶ For two finite sets, this is easy, just count the number of elements and compare them!
- ▶ But what about two infinite sets?
- ▶ Example: {set of all even natural numbers} vs $\mathbb{N}$ vs $\mathbb{Q}$ vs $\mathbb{R}$
- ▶ Turns out we need functions... but first...

Hilbert's hotel



- ▶ Suppose there is a hotel with infinitely many rooms.
 - ▶ And suppose they are all full (like in IIT guest house).
1. Can you accomodate 1 or finitely many more guests, by shifting around the existing guests?
 2. What if infinitely many more guests arrive?
 3. What if infinitely many more trains with infinitely many more guests arrive?
- (H.W)

Functions

What you did above was to define functions...

Definition

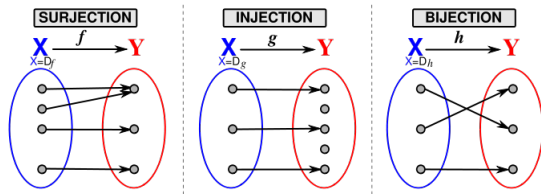
Let A, B be two sets. A **function** f from A to B is an assignment of exactly one element of B to each element of A . $f : A \rightarrow B$ is a **subset R of $A \times B$** such that

- (i) $\forall a \in A, \exists b \in B$ such that $(a, b) \in R$, and
- (ii) if $(a, b) \in R$ and $(a, c) \in R$, then $b = c$.

Inverse of a function

- If $f : A \rightarrow B$ is a **??? function**, then $f^{-1} : B \rightarrow A$ defined by $f^{-1}(b) = a$ if $f(a) = b$, is called its inverse.
- If f is a bijection, then $f \circ f^{-1} = f^{-1} \circ f = \text{id}$

Comparing (finite and infinite) sets



- ▶ **Surjective or onto:** $f : A \rightarrow B$ is surjective if $\forall y \in B, \exists x \in A$ such that $f(x) = y$.
 - If A, B finite, $|A| \geq |B|$
 - ▶ **Injective or 1-1:** $f : A \rightarrow B$ is injective if $\forall x, y \in A$, if $f(x) = f(y)$, then $x = y$.
 - If A, B finite, $|A| \leq |B|$
 - ▶ **Bijective or 1-1 correspondence:** A function is bijective if it is surjective and injective.
 - If A, B finite, $|A| = |B|$
1. $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2$.
 2. $f : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ such that $f(x) = x^2$.

Properties of finite and infinite sets

Similarities between finite and infinite sets

- ▶ $\exists$ **bij** from A to B and B to C , implies $\exists$ **bij** from A to C .
- ▶ $\exists$ **bij** from A to B , then $\exists$ **bij** from B to A .
- ▶ (**Schröder-Bernstein Theorem:**) $\exists$ **inj** from A to B and $\exists$ **inj** B to A , implies $\exists$ **bij** from A to B . (Read this proof!)

Differences between finite and infinite sets

- ▶ For finite sets, if A is a set and $b \notin A$, then $|A| \neq |A \cup \{b\}|$.
- ▶ What about infinite sets?